

DATA ANALYTICS PROJECT

Northwind Sales Analytics

A complete guide to analysing customer behaviour, order patterns, and operational performance across six key business dimensions



OVERVIEW

Project Scope and Objectives

What We're Analysing

This project examines six critical business areas: customer purchasing patterns, order performance, employee productivity, product inventory, supplier relationships, and shipping efficiency. By connecting data across multiple tables, we uncover actionable insights that drive strategic decisions.

Expected Outcomes

Identify high-value customers and churn risks, optimise inventory levels, improve delivery performance, enhance workforce planning, and strengthen supplier partnerships. These insights support marketing, sales, and operational improvements across the organisation.

Six Dimensions of Business Intelligence



Customer Analysis

Understanding purchasing behaviour, identifying high-value customers, and tracking order frequency to reduce churn



Order Analysis

Evaluating sales performance, tracking revenue trends, and optimising fulfillment processes



Employee Analysis

Examining workforce distribution, tenure patterns, and workload allocation



Product Analysis

Assessing inventory levels, pricing strategies, and product performance across categories



Supplier Analysis

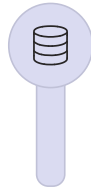
Evaluating supplier reliability, geographic distribution, and procurement efficiency



Shipper Analysis

Monitoring delivery performance, shipment volume, and logistics optimisation

Six-Step Implementation Process



Understand the Dataset

Review all tables including Customers, Orders, OrderDetails, Products, Employees, Suppliers, Shippers, and Categories. Map relationships and identify data quality issues.



Data Preparation

Clean data using Excel and SQL. Remove duplicates, fix formatting inconsistencies, handle missing values, and create calculated columns for analysis.



Data Analysis

Execute SQL queries to summarise key metrics. Calculate total revenue, average order values, customer segments, and product performance indicators.



Power BI Modeling

Import cleaned datasets, establish table relationships, and create DAX measures for revenue, customer lifetime value, and performance metrics.



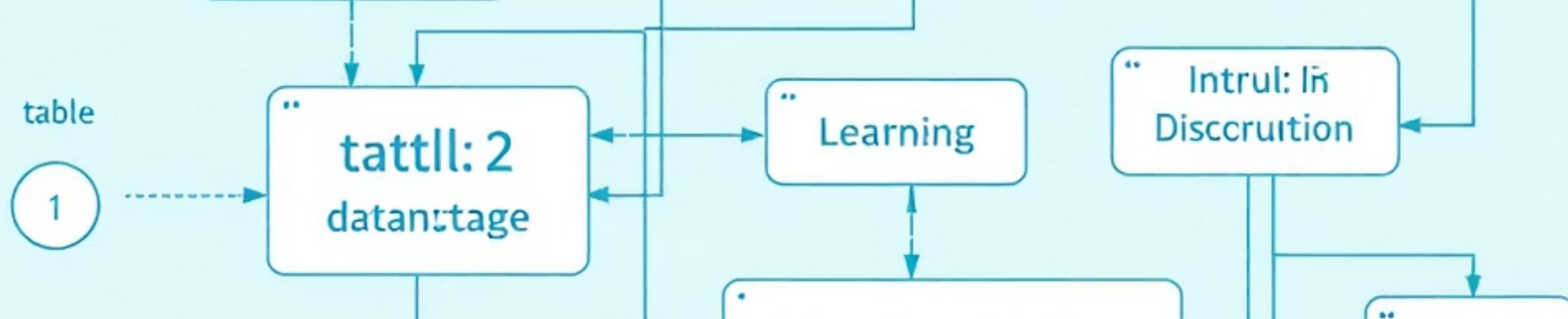
Visualisation

Build interactive dashboards with charts, tables, and KPIs. Display trends, comparisons, and performance metrics across all analysis dimensions.



Insights & Reporting

Summarise findings and provide actionable recommendations for marketing strategies, sales optimisation, and operational improvements.



STEP 1-2

Data Foundation: Understanding and Preparation

Dataset Review

Begin by mapping all table relationships. OrderID connects Orders to OrderDetails, ProductID links Products to OrderDetails, and CustomerID ties Orders to Customers. Check for missing data and duplicates.

Excel Preparation

Import all CSV files, remove duplicates, standardise formatting, and handle null values. Create calculated columns such as Total_price using $\text{Quantity} \times \text{UnitPrice}$ adjusted for discounts.

SQL Foundation

Load datasets into database tables with proper primary and foreign keys. Write queries to clean data, remove duplicates, and create aggregated tables for revenue by customer, product, and region.

Analysis and Modelling Excellence

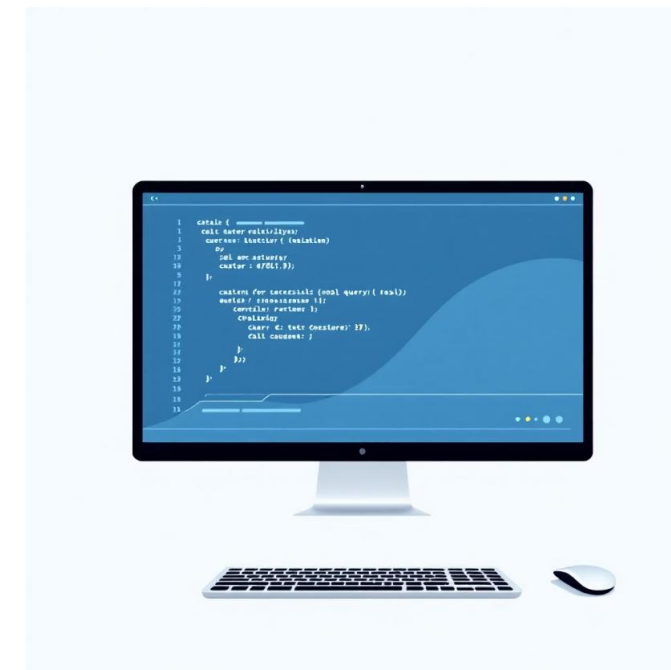
SQL Analysis Queries

Execute comprehensive queries to explore and summarise the dataset:

- Calculate total revenue, order count, and average order value
- Determine orders per customer and identify top-selling products
- Measure employee performance through orders handled and tenure analysis
- Assess supplier and shipper reliability metrics

Power BI Data Modeling

Create relationships between imported tables and build DAX measures including Total Revenue, Average Order Value, and Customer Lifetime Value calculations for deeper insights.



Visualisation Strategy



Customer Dashboards

Display total revenue, top customers by spend, order frequency distribution, and customer lifetime value segments with trend analysis



Operations Metrics

Track employee order handling, tenure statistics, supplier product counts, shipping performance, and delivery time analysis

Utilise bar charts, line graphs, pie charts, detailed tables, and prominent KPI cards to create an intuitive, actionable dashboard experience.



Order & Product Views

Show monthly and quarterly trends, average order values, discount impact, top-selling products, stock levels, and category performance

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Business Intelligence

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KEY INSIGHTS

Expected Analytical Discoveries

Customer Intelligence

Identify your most valuable customers, understand purchase frequency patterns, and detect early warning signs of potential churn risks

Product Performance

Discover best-selling items, optimise inventory levels, manage stock efficiently, and refine pricing strategies based on demand

Workforce Efficiency

Highlight employees with highest order processing rates, balance workload distribution, and support performance evaluation processes

Supply Chain Excellence

Evaluate supplier reliability, assess shipper performance, reduce delivery times, and strengthen procurement relationships

RECOMMENDATIONS

Actionable Business Strategies



Marketing Optimisation

Target high-value customers with personalised campaigns. Re-engage at-risk customers with retention programmes. Leverage purchase patterns for cross-selling opportunities.



Sales Enhancement

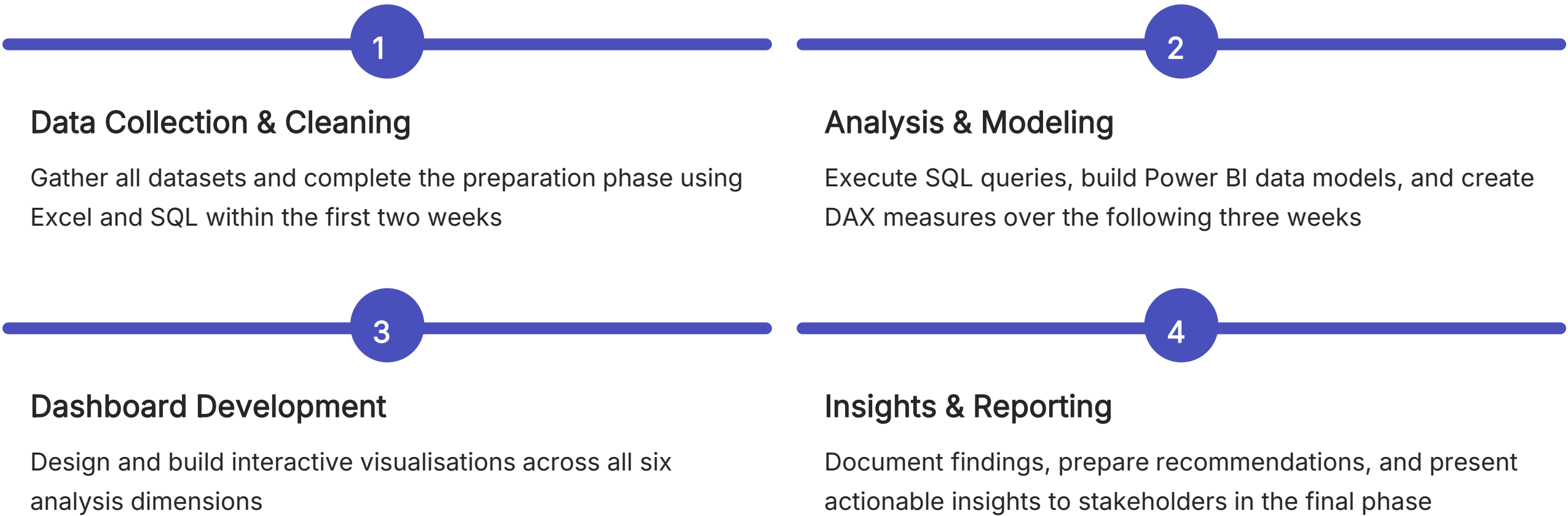
Focus efforts on top-performing products and regions. Adjust pricing strategies based on demand analysis. Develop promotions around seasonal trends.



Operations Excellence

Optimise inventory to prevent stockouts and overstock. Improve employee training based on performance gaps. Strengthen partnerships with reliable suppliers and shippers.





1

Data Collection & Cleaning

Gather all datasets and complete the preparation phase using Excel and SQL within the first two weeks

2

Analysis & Modeling

Execute SQL queries, build Power BI data models, and create DAX measures over the following three weeks

3

Dashboard Development

Design and build interactive visualisations across all six analysis dimensions

4

Insights & Reporting

Document findings, prepare recommendations, and present actionable insights to stakeholders in the final phase